INTERNATIONAL MARITIME ORGANIZATION

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SUB-COMMITTEE ON NAVIGATION, COMMUNICATIONS AND SEARCH AND RESCUE
13th session
Agenda item 3

NCSR 13/3/7
20 March 2026
Original: ENGLISH
Pre-session public release: ☒

ROUTEING MEASURES AND SHIP REPORTING SYSTEMS

Amendments to the traffic separation scheme "In the approaches to Chesapeake Bay"

Submitted by the United States

SUMMARY

Executive summary:	This document contains a proposal for the establishment of a new precautionary area associated to the traffic separation scheme *In the approaches to Chesapeake Bay*. This proposed amendment is intended to maintain traditional shipping routes and continue to ensure that these navigation lanes remain free of fixed structures.
Strategic direction, if applicable:	7
Output:	7.22
Action to be taken:	Paragraph 30
Related documents:	*Ships' Routeing*, 2019 edition; SOLAS regulation V/10; resolution A.572(14), as amended; MSC.1/Circ.1060, as amended; COLREG.2/Circ.55; NCSR 13/3/4, NCSR 13/3/5, NCSR 13/3/6 and NCSR 13/INF.9

Introduction

1 The United States proposes to amend the traffic separation scheme (TSS) *In the approaches to Chesapeake Bay* adopted originally by IMO on 12 October 1971 and last amended on 15 December 2004. The existing TSS consists of three parts: a precautionary area, an eastern approach and a southern approach. The southern approach consists of inbound and outbound lanes separated by a deep-water (DW) route for inbound and outbound vessels with draughts exceeding 12.8 metres (42 feet) in fresh water and for naval aircraft carriers.

Summary

2 This document proposes to establish a new precautionary area connecting the offshore termini of the eastern and southern approaches to the TSS *In the approaches to Chesapeake Bay*. This proposed precautionary area would be approximately 22 nautical miles (NM thereafter) long and 5 NM wide, bounded by arcs of 5 NM. Figure 1 provides a side-by-side comparison of the current TSS with the proposed amendments.

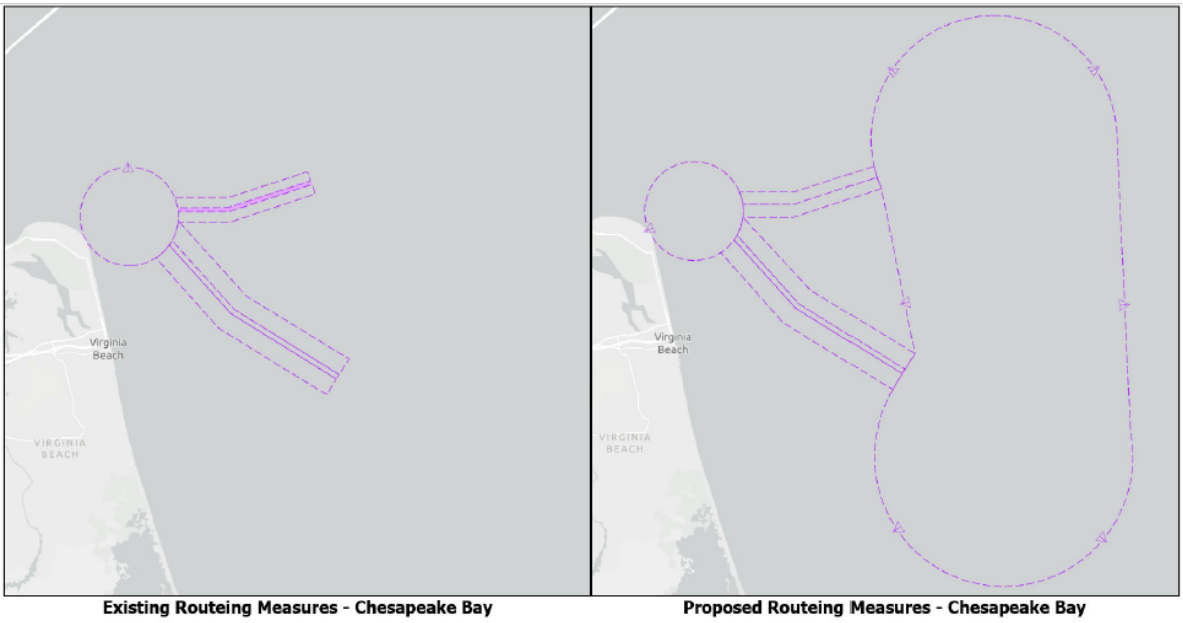

Figure 1: Proposed amendments to TSS *In the approaches to Chesapeake Bay*

3 The United States has placed particular attention on expanded oil, natural gas, hydrokinetic, solar, nuclear, aquaculture and other non-mineral energy resources on the outer continental shelf (OCS). In identifying viable locations in the United States exclusive economic zone (EEZ) for energy production, the United States has also identified safe access routes, or "fairways", for the movement of vessel traffic proceeding to or from United States ports, in which no artificial island or fixed structure, whether temporary or permanent, will be permitted. Figure 3 shows the United States fairways in the approaches to Chesapeake Bay depicted in green. As such, the designation of a fairway recognizes the paramount right of navigation over all other uses in the designated areas. Based on the historical shipping trends provided in paragraphs 10 and 11, coupled with the planned construction of offshore structures (depicted in blue and black in figure 3), the United States expects shipping transits in the approaches to Chesapeake Bay to consolidate into these designated fairways. Multiple shipping routes, as seen in figure 4, will converge at the eastern and southern approaches to the TSS *In the approaches to Chesapeake Bay* at the proposed locations.

4 Following a vessel collision in 1998, the United States established a "Manoeuvring Area" between the eastern and southern approaches as seen in figure 2. While charted and used like a precautionary area, this area was not proposed as a precautionary area to IMO as it was initially established with the intent to address a local navigational safety concern resulting from a collision involving vessels conducting calibration tests in the manoeuvring area between the TSS. With the expected growth of offshore energy construction projects and the subsequent consolidation of vessel traffic, which includes a significant number of internationally flagged vessels, into confined routes in the area, the United States believes it is prudent to include this area as part of the proposed precautionary areas.

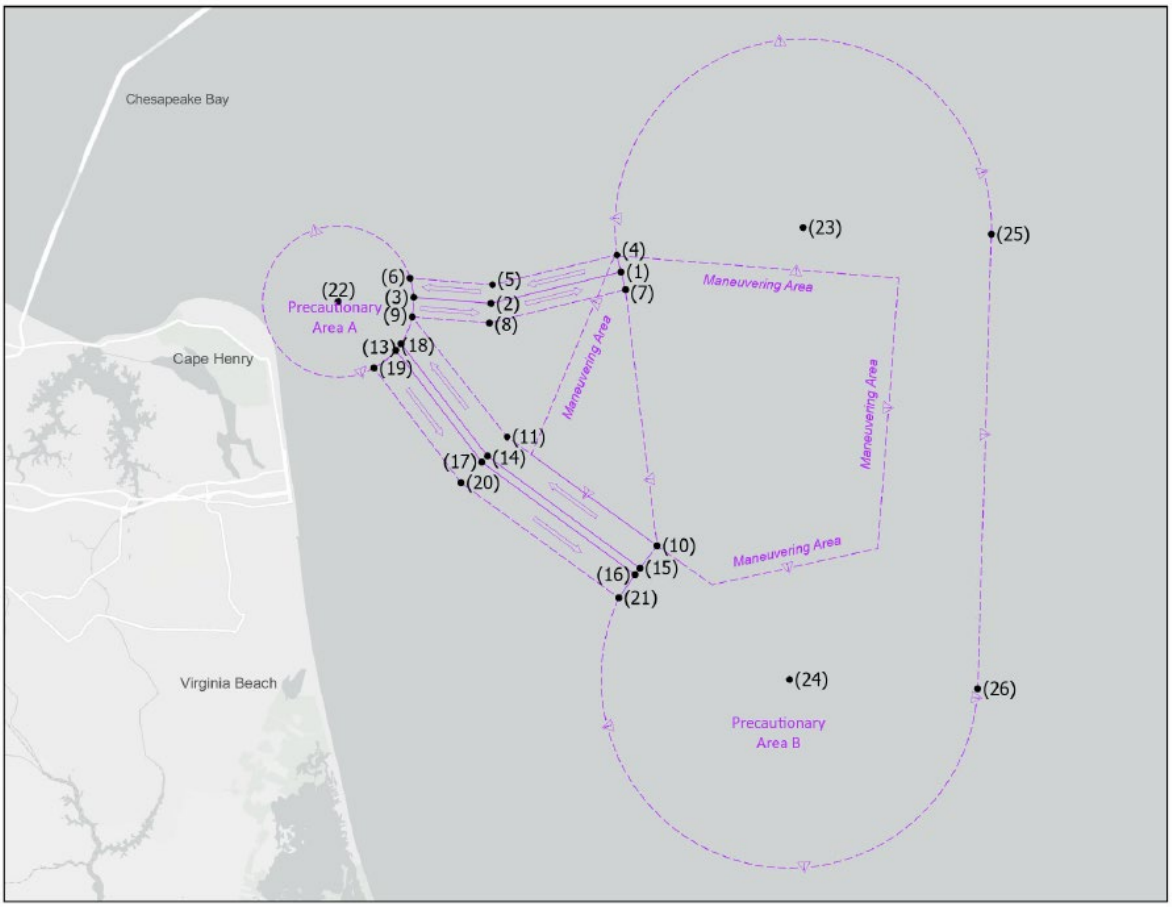

Figure 2: Chesapeake Bay manoeuvring area

5 A precautionary area is the preferred routeing measure as it provides a charted identifiable area where ships must navigate with particular caution. A direction of traffic flow was considered, but determined to be unnecessary and inconsistent with all other precautionary areas throughout the United States. This proposal is preventative in nature and will help provide predictable navigation corridors to accommodate shipping routes along with the evolving offshore energy infrastructure, thereby improving operational safety in increasingly congested areas of Chesapeake Bay. This system will continue to apply to all ships transiting the area. All other existing schemes remain unchanged. Further discussion on traffic considerations is provided in paragraphs 10 and 11.

6 In addition, this document presents all geographical coordinates in the IMO *Ships' Routeing* 2019 edition description of the TSS *In the approaches to Chesapeake Bay* converted from the existing North American 1983 geodetic datum to the World Geodetic System of 1984 datum. Updated coordinates that have changed with respect to the shift in datum are provided in the annex.

7 A study, Port Access Route Study: Approaches to Chesapeake Bay, is referenced in document NCSR 13/INF.9. The study aimed to characterize whether amendments to existing, or additional routeing measures are necessary to improve navigation safety due to factors such as planned or potential offshore energy development, current port capabilities and planned improvements, increased vessel traffic, existing and potential anchorage areas, changing vessel traffic patterns, meteorological and oceanographic effects, or navigational difficulty in the proposal area. An in-depth analysis of the history of groundings, collisions, or damage to the marine environment can be found in section E of the study. There has been no significant growth in these incidents that is precipitating the establishment of these precautionary areas.

This proposal is submitted in advance of offshore construction as a preventative measure to ensure the safety of navigation in the area.

Description of the area

8 The location of the proposed precautionary areas, existing TSS, the United States fairways, and offshore energy areas are provided in figure 3. Geographic positions, number, edition and geodetic datum of the reference chart used to define the amended routeing system, are set out in the annex.

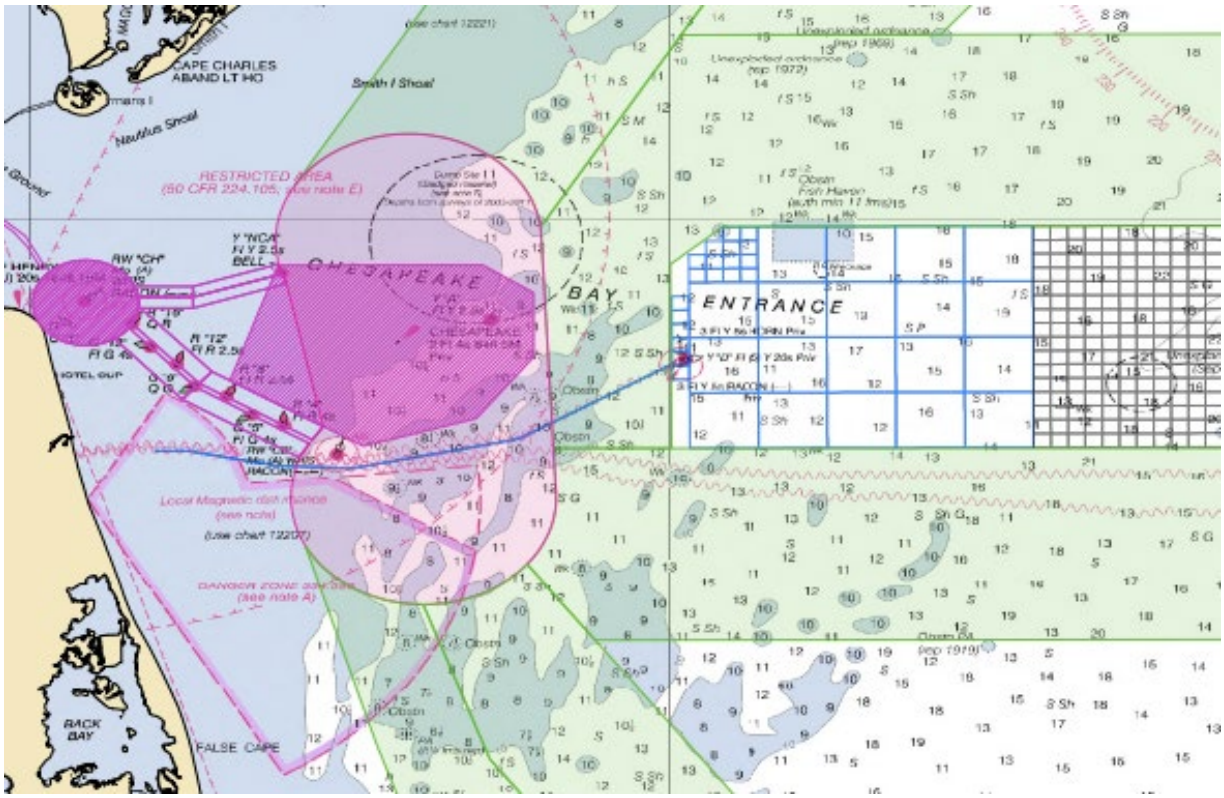

Figure 3: Description of area

Cooperation between interested States

9 The United States has rights, jurisdiction and obligations to the waters in the proposal to amend the TSS *In the approaches to Chesapeake Bay*. The TSS is entirely in the exclusive economic zone (EEZ) of the United States and there is no need for a cooperation agreement with other States.

Traffic considerations

10 The United States analysed vessel traffic in the approaches to Chesapeake Bay and focused on waters located seaward of existing port approaches within the United States EEZ. The study referenced in document NCSR 13/INF.9 allowed the United States to consider vessel movements among both domestic and international ports of call to inform marine planning in the approaches to Chesapeake Bay. Information used included automatic identification system (AIS) data and information from shipping organizations to identify traditional navigation routes. The AIS data identified customary routes transited by vessels. Because these vessel types have different manoeuvring capabilities and navigation safety requirements, the identification of their customary routes and required shipping route widths were studied separately and aggregated for the final proposal.

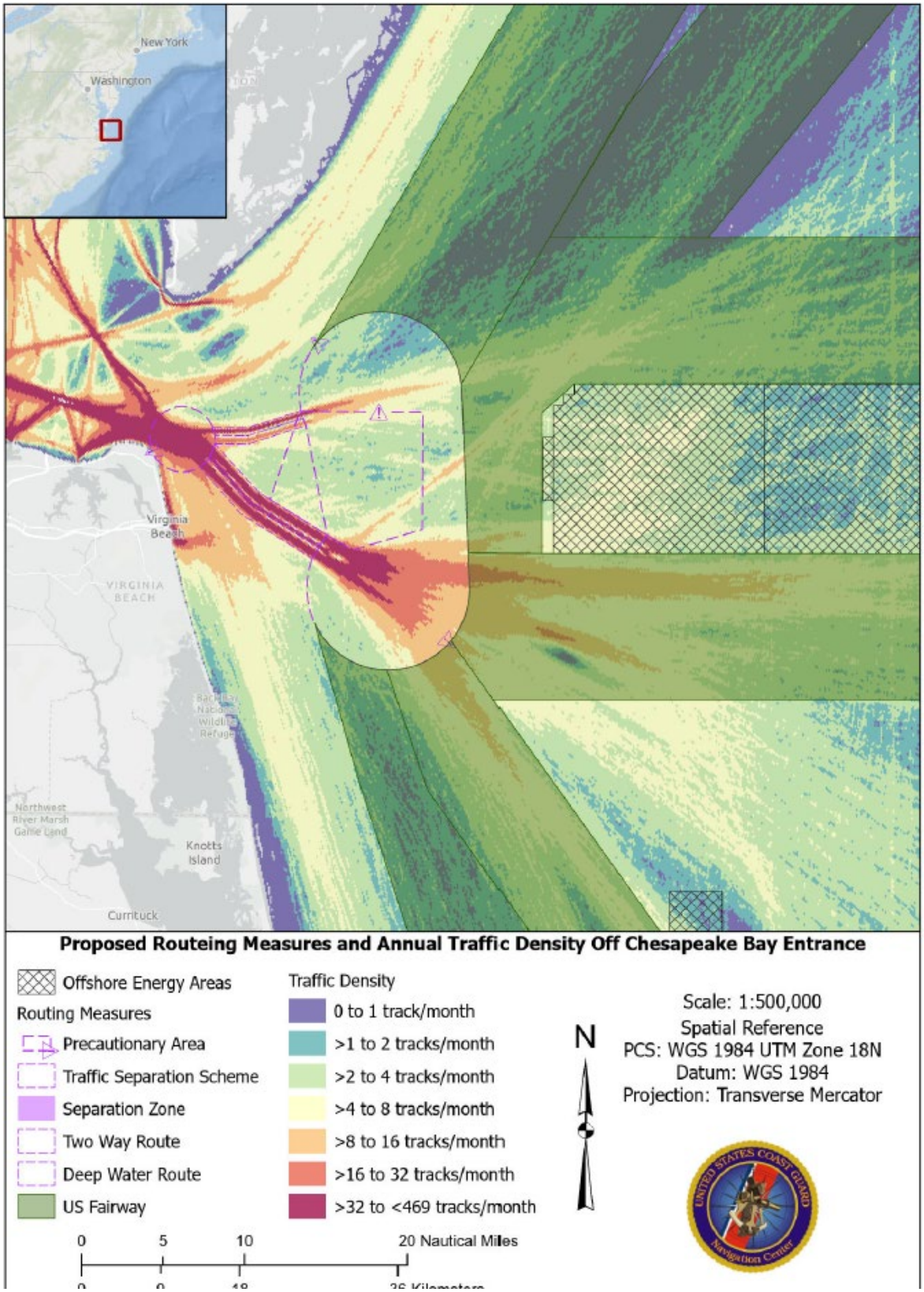

Figure 4: AIS vessel movements in the approaches to Chesapeake Bay

11 While AIS transits in red in figure 4 indicate predominant use of the established routeing measures, the additional traditional shipping routes shown in blue, yellow and green indicate a significant amount of vessel traffic that historically transits through the offshore energy construction sites seen in black. Additionally, note the higher concentration of northeast/southwest vessel tracks extending from the southern approach through the middle of the manoeuvring area and proposed precautionary area. These depict the deep draught ships (>12.8 metres/42 feet) entering and exiting Chesapeake Bay through the established deep-water route. There is a naturally occurring deep-water slough that extends through the manoeuvring area and proposed precautionary area and to the north of the offshore energy area. By establishing the United States fairways as described in paragraph 3, the United States, in consultation with the commercial shipping industry, expects to see a consolidation of shipping traffic into the defined United States fairways surrounding the proposed precautionary area. By consolidating traffic into these fairways, shipping transits in the area will converge with the TSS in the proposed precautionary area. The consolidation of shipping, the necessary safe track for deep draught vessels, and the manoeuvring area make the proposed precautionary area a necessity for safe navigation.

12 The existing aids to navigation (AtoN) can be seen in figure 5.

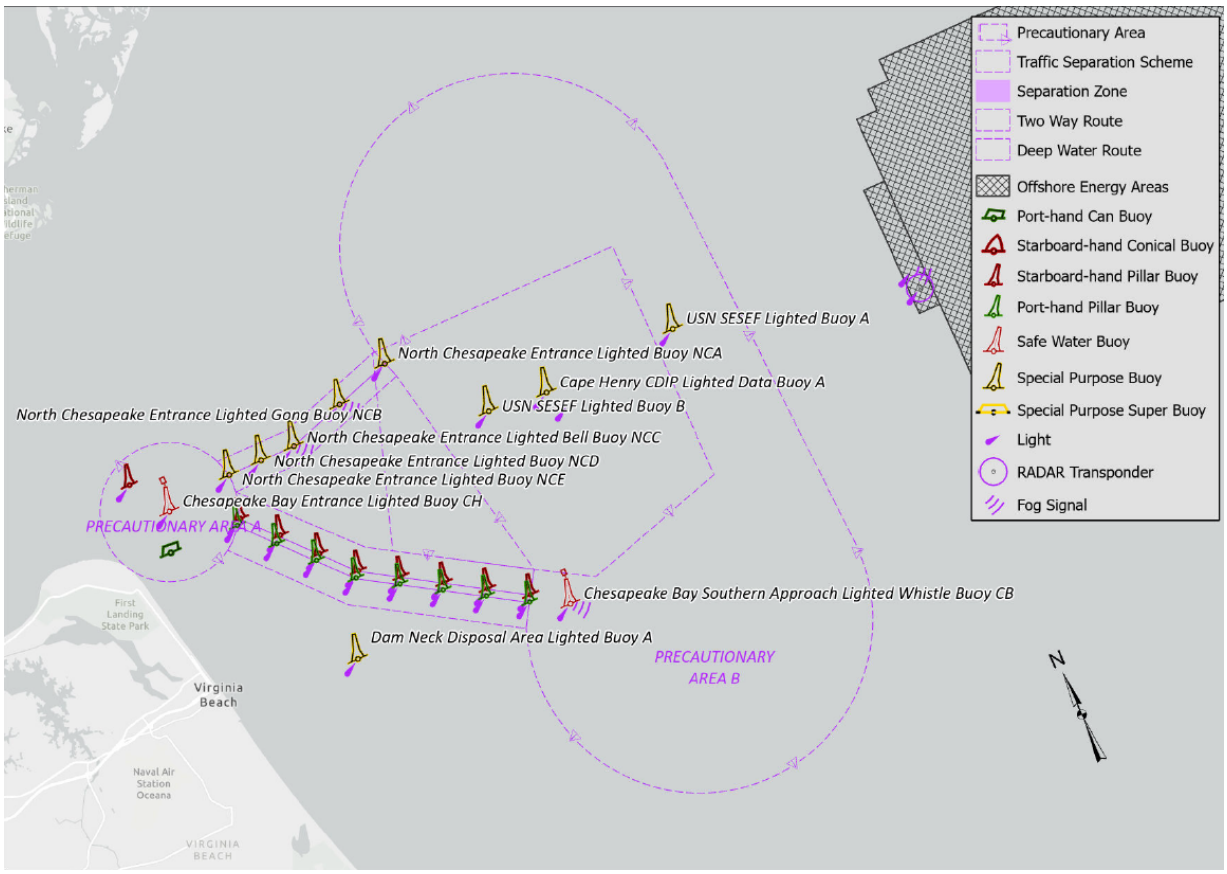

Figure 5: Existing aids to navigation in the approaches to Chesapeake Bay

Eastern approach

13 The eastern approach is marked by five special (yellow) buoys in the centre of the separation zone. These buoys are approximately 1 to 1.5 NM apart, labelled, from offshore to inshore, "NCA", "NCB", "NCC", "NCD" and "NCE". Each buoy is lighted with a distinct yellow flash characteristic of 2.5 seconds ("NCA"), 6 seconds ("NCB"), 4 seconds ("NCC"), 2.5 seconds ("NCD"), and 6 seconds ("NCE"). The United States does not intend to relocate these buoys upon adoption of this proposal as they support the eastern approach and do not conflict with the proposed amendment.

Southern approach

14 A deep-water route is established in the separation zone of the southern approach. A safe water buoy labelled "CB" with a white morse "A" flash characteristic of 8 seconds is positioned 1 NM to the southeast of the offshore terminus of the southern approach. This buoy also provides a morse "A" RACON. A series of eight gated pair, lighted buoys starting at the offshore terminus of the TSS, spaced 1.25 NM apart, mark the deep-water route. The United States does not intend to relocate these buoys upon adoption of this proposal as they support the southern approach and do not conflict with the proposed amendment.

Precautionary area A

15 A safe water buoy labelled "CH" with a white morse "A" flash characteristic of 8 seconds is positioned in the centre of the existing precautionary area. The "CH" buoy also provides a morse "A" RACON and marks a pilot boarding area. These AtoN do not conflict with the proposed amendments, and the United States does not intend to relocate upon adoption of this proposal.

16 Several United States Navy buoys are located in and around the Manoeuvring Area discussed in paragraph 4. Dam Neck Disposal Area Light Buoy "A" is located to the south of the southern approach. These buoys are not used for navigation purposes and the United States does not intend to relocate upon adoption of this proposal.

17 Virtual AtoN in the proposed area are under consideration as safe water marks off the Eastern and Southern approaches. If a virtual AtoN is established in the southern approach, the physical "CB" buoy may be removed.

18 There is no Vessel Traffic Service in the area.

19 The United States National Oceanic and Atmospheric Administration (NOAA) hydrographic surveys for US4VA1AG, 2025 edition; US4VA1BG, 2024 edition; US4VA1AH, 2025 edition; US4VA1BH, 2025 edition; US5VA1EV, 2024 edition; and US5VA1FV, 2024 edition adhere to International Hydrographic Organization (IHO) standards, including special publication 44. These surveys accurately represent the water depths in the area, and the depths provided on the nautical charts were updated in January 2025. All paper charts in the United States were discontinued by January 2025. Corresponding ENCs are provided and updated in the annex as appropriate.

20 Future offshore energy development is expected along the United States central Atlantic coast. The United States Bureau of Ocean Energy Management (BOEM) has published a proposed "Call Area", which consists of 13,476,805 acres (54,539 square kilometres) off the United States central Atlantic coast (figure 6). While the leased construction sites displayed in figures 3 and 4 show an already agreed upon area for offshore energy production, additional sites could be leased inside the Call Area.

Adoption of this proposal along with the establishment of United States fairways, as discussed in paragraph 3, will prohibit construction in shipping lanes and ensure safe passage to and from United States ports.

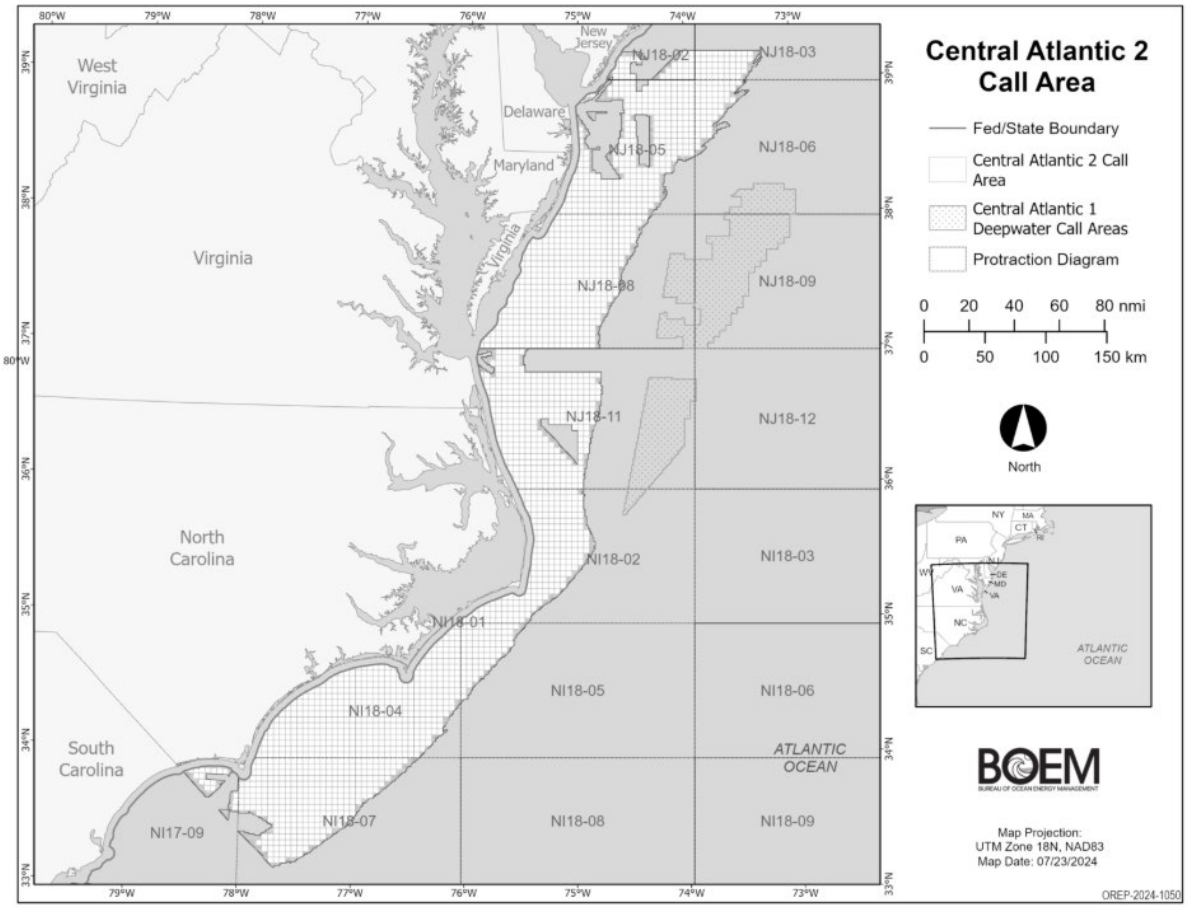

Figure 6: Proposed US offshore energy areas

Marine environmental considerations

21 The prevailing wind directions in the area are along the coast during the summer months and across the coast during the winter months. Average wind speed is 14-18 knots with seas of 2-3 feet. The strongest winds are observed during the presence of a hurricane off the coast of the United States Atlantic; the easterly cyclonic winds along the northern edge of the storm could drive a strong surge to the west, laterally along the southern coast of Long Island. A few times per year, primarily in the winter months, the wind direction is north-east to north.

22 The United States considered vessel noise, vessel operations and vessel movements in the area of the proposal. These considerations are not expected to result in significant impacts to the following resources: air quality, ambient sound, marine vegetation, marine invertebrates, flying insects, birds, bats, marine fish, essential fish habitat, commercial fishing, marine construction, mineral extraction, oil and gas extraction, recreation and tourism, existing energy projects, research, transportation and shipping, and subsistence fishing and hunting.

23 There are no marine environmental considerations that will impact this proposal.

24 Additional information regarding marine and environmental considerations can be found in document NCSR 13/INF.9.

Status of the routeing system

25 This proposal is recommendatory and is designed to preserve traditional maritime commerce routes and safe access to United States ports and protect them from fixed structures that could impact navigation safety and generally prioritize the right of navigation over all other uses. By establishing the United States fairways as described in paragraph 3, the United States, in consultation with the commercial shipping industry, expects to see a consolidation of shipping traffic into the defined fairways seen in green in figures 1 and 3. By consolidating traffic into these fairways, both port entry and coastwise shipping transits will converge with the TSS in the proposed precautionary area.

Position-fixing in relation to the routeing system

26 Accurate position-fixing is widely available to all vessel traffic using the Global Positioning System (GPS). Aids to navigation, including the establishment of virtual AtoN, are addressed in paragraphs 12 through 17. Because of low user numbers and significant operational costs compared to satellite-based GPS, LORAN-C was considered unnecessary for modern navigation and discontinued in February 2010. Nautical charts for the area are available. GPS and nautical charts used in conjunction with AIS provide real-time location data to other vessels and shore stations.

Miscellaneous information

27 Prior to establishing the fairways, as discussed in paragraph 3, the United States conducted a Port Access Route Study (PARS) to determine traffic density; if current vessel routeing measures are adequate or require modifications; the type of new vessel routeing measures as necessary; and if the usage of the vessel routeing measures should be mandatory for specific classes of vessels. As provided in paragraph 7, a copy of this study is provided in document NCSR 13/INF.9.

28 Mariner participation in a PARS is essential for the United States to gather firsthand operational data, ensuring that proposed, updated, or new fairways and routeing measures safely accommodate vessel traffic while mitigating risks from offshore development and congestion. Chesapeake Bay pilots, local port authorities, commercial fishing, passenger vessel operators, bulk and container vessel operators and recreational boaters were consulted throughout the study to capture direct input regarding the proposed area. Pertinent information has been provided in this proposal. Additional information captured from the consultations and subsequent analysis is provided in document NCSR 13/INF.9.

Proposed date of implementation

29 It is proposed that the amendment enter into force six months after adoption by the Committee.

Action requested of the Sub-Committee

30 The Sub-Committee is invited to consider the proposal in the annex and recommend it to the Committee for adoption.

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ANNEX

PROPOSED AMENDED TRAFFIC SEPARATION SCHEME AND ASSOCIATED MEASURES "IN THE APPROACHES TO CHESAPEAKE BAY"

The highlighted areas are the proposed changes.

IN THE APPROACHES TO CHESAPEAKE BAY

(Reference charts: ~~United States 1220, 2002 edition, 12207, 1998 edition; 12221, 2003 edition.~~ US4VA1AG, 2025 edition; US4VA1BG, 2024 edition; US4VA1AH, 2025 edition; US4VA1BH, 2025 edition; US5VA1EV, 2024 edition; US5VA1FV, 2024 edition.
Note: These charts are based on ~~North American 1983~~ World Geodetic System of 1984 datum)

Description of the traffic separation scheme

The traffic separation scheme *In the approaches to Chesapeake Bay* consists of three parts:

Part I

Precautionary areas

(a) Precautionary area A of radius two-mile is centred upon geographical position (22) 36°56′.13 N, 075°57′.45 W.

(b) Precautionary area B is established bounded by a line connecting the following geographical positions from:

(4) 36°57′.95 N, 075°48′.37 W;
then easterly by an arc of 5 NM centred at (23) 36°59′.06 N, 075°42′.28 W to
(25) 36°59′.27 N, 075°36′.04 W;
then southerly to (26) 36°47′.20 N, 075°35′.35 W;
then westerly by an arc of 5 NM centred around (24) 36°47′.06 N, 075°41′.57 W to
(21) 36°48′.87 N, 075°47′.42 W;
then north-easterly to (10) 36°50′.33 N, 075°46′.29 W;
then north-westerly to (7) 36°57′.04 N, 075°47′.99 W;
then north-westerly to (4) 36°57′.95 N, 075°48′.37 W.

Part II

Eastern approach

(a) A separation line connects the following geographical positions:

(1) 36°57′.5~~0~~1 N, 075°48′.~~21~~18 W (3) 36°56′.40 N, 075°54′.9~~5~~7 W
(2) 36°56′.40 N, 075°52′.40 W

(b) A traffic lane for westbound traffic is established between the separation line and a line connecting the following geographical positions:

(4) 36°57′.9~~4~~5 N, 075°48′.~~41~~37 W (6) 36°56′.90 N, 075°55′.14 W
(5) 36°56′.90 N, 075°52′.40 W

(c) A traffic lane for eastbound traffic is established between the separation line and a line connecting the following geographical positions:

(7) 36°57′.04 N, 075°~~48′.01~~47′.99 W (9) 36°55′.88 N, 075°54′.9~~5~~7 W
(8) 36°55′.88 N, 075°52′.40 W

Part III

Southern approach

(a) A separation line connects the following geographical positions:

(10) 36°50′.33 N, 075°46′.29 W (12) 36°55′.~~96~~88 N, 075°54′.97 W
(11) 36°52′.90 N, 075°51′.52 W

(b) A separation line connects the following geographical positions:

(13) 36°55′.1~~4~~5 N, 075°55′.2~~3~~7 W (15) 36°49′.70 N, 075°46′.80 W
(14) 36°52′.35 N, 075°52′.12 W

(c) A separation line connects the following geographical positions:

(16) 36°49′.52 N, 075°46′.94 W (18) 36°54′.9~~7~~6 N, 075°55′.4~~3~~2 W
(17) 36°52′.18 N, 075°52′.29 W

(d) A separation line connects the following geographical positions:

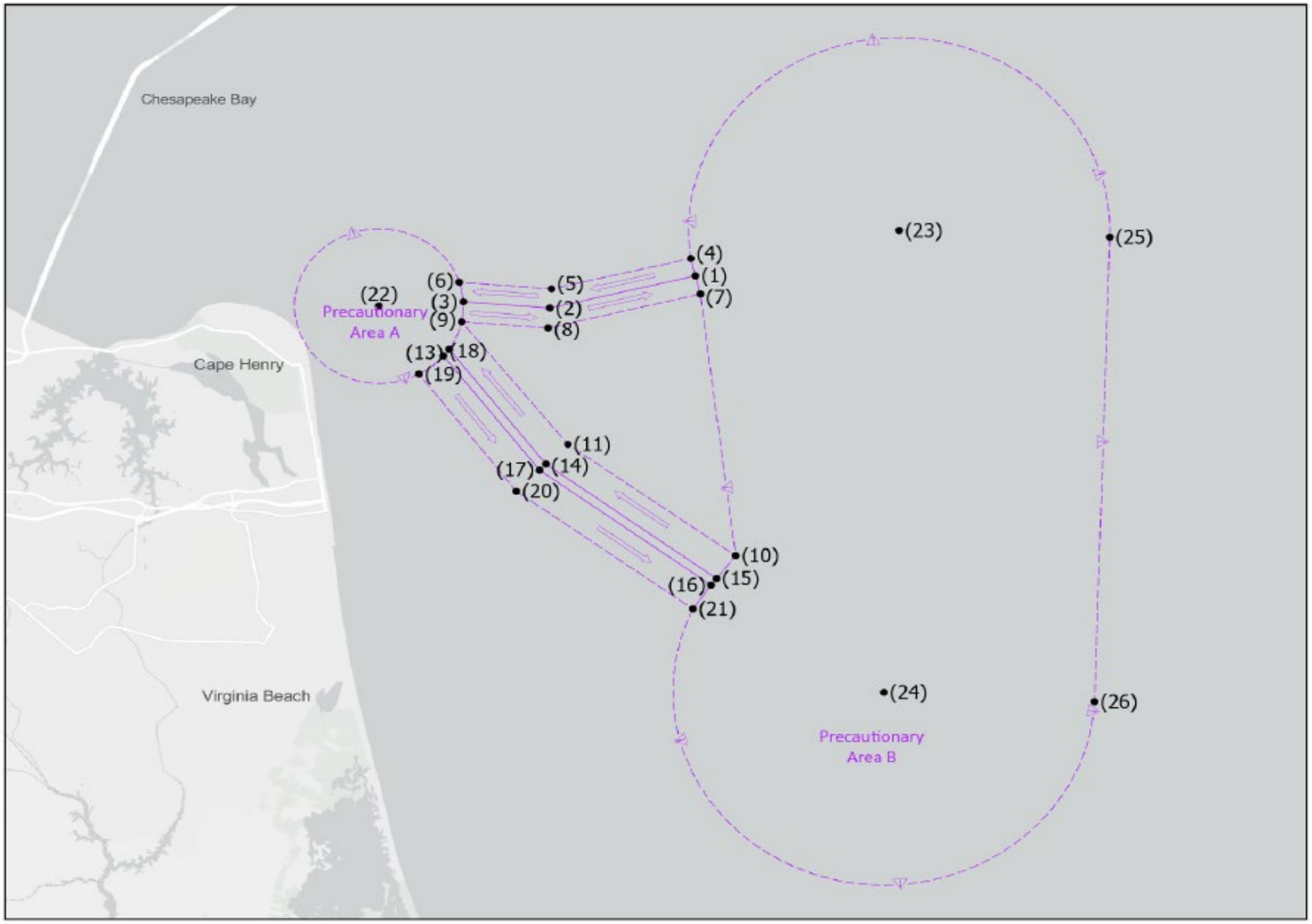
(19) 36°54′.44 N, 075°56′.09 W (21) 36°48′.87 N, 075°47′.42 W
(20) 36°51′.59 N, 075°52′.92 W

(e) A traffic lane for inbound traffic is established between the separation lines described in paragraphs (a) and (b).

(f) A traffic lane for outbound traffic is established between the separation lines described in paragraphs (c) and (d).

Deep-water route

(g) A deep-water route is established between the separation lines described in paragraphs (b) and (c). The types of ships which are recommended to use the deep-water route are given in the description of the deep-water route (see part C, section IV). All other ships using the southern approach traffic separation scheme should use the appropriate inbound or outbound traffic lane.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